

List of programs for OS Lab External

B.Tech IV SEM CSE-4

Date of Exam :19-06-2026

Part-1

1. Write a C program to simulate the Paging memory management technique
2. Write C program to illustrate FIFO IPC mechanism.
3. Write a C program to implement the Producer – Consumer problem using semaphores
4. Write a C program to illustrate Shared Memory IPC mechanism.
5. Write programs using the I/O system calls of UNIX/LINUX operating system (open, read, write, close, stat, opendir, readdir)
6. Write a C Program to illustrate ordinary pipes.
7. Write a C program to Implement FCFS CPU Scheduling algorithm without arrival time.
8. Write a C program to Implement SJF CPU Scheduling algorithm without arrival time.
9. Write a C program to Implement Priority CPU Scheduling algorithm without arrival time

Part-2

1. Write C programs to simulate the Segmentation memory management technique.
2. Write a C Program to illustrate Message Queues IPC mechanism.
3. Write a C Program to implement FIFO Page replacement Algorithm.
4. Write a C Program to implement LRU Page Replacement Algorithm.
5. Write a C Program to implement Optimal Page Replacement Algorithm.
6. Write a C Program to implement round robin CPU Scheduling Algorithm
7. Write a C program to Implement FCFS CPU Scheduling algorithm with arrival time.
8. Write a C program to Implement SJF CPU Scheduling algorithm with arrival time.
9. Write a C program to Implement Priority CPU Scheduling algorithm with arrival time.
10. Write a C program to simulate Bankers Algorithm for Deadlock Avoidance and Prevention.

Theory Questions

- 1.Critical Section Problem and Petersons solution**
- 2.Fixed and dynamic Partitioning(contiguous allocation)**
- 3.Free space management.**
- 4.Dining Philosophers using Monitors**
- 5.TLB.**
- 6.Reader -Writers Problem using Semaphores.**
- 7.Usage of lseek,stat,iocctl system calls.**
- 8.NTFS and FAT File system**
- 9.Deadlock Detection and recovery**
- 10.System calls.**
- 11.Schedulers.**
- 12.OS Services**
- 13.Os Components and its activities**
- 14.OS Structures**
- 15.Messagepassing ,shared Memory**

Instructions

- 1)Every student has to come in formals(no jeans,no jeggins,no cotton jeans,no sleeveless)**
- 2)Every student should wear Id cards before entering the lab.**
- 3)Every Student should carry corrected observation and record (certified)**
- 4)No exchange of questions(2 programs is mandatory and theory question)**
- 5) No student is allowed to carry mobile phone,smart watches,USB any gadgets to exam.**

6)Late commers will not be allowed to exam(Batch-1:9:15 AM,Batch-2:11:30 AM)

7)Every one should write (question,Aim,Alogorithm,Program,Expected input output)

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